

Maneuvering Against a Directional Field: Path-Integrated Exposure, Commitment, and Delayed Threats in Nonholonomic Adversarial Games

Research Layer of the *Iron Bottom Sound* project*

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Abstract

We study adversarial trajectory optimization between two moving *formations* that interact through a directional field under nonholonomic motion constraints. The decision variable is not a position but a trajectory, and the objective is the path-integrated net exposure $J = \int_0^T [P_{B \rightarrow R}(t) - \lambda P_{R \rightarrow B}(t)] dt$ built from a calibrated anisotropic interaction kernel; we characterize the resulting closed-loop dynamic game and certify its comparative statics numerically. Three structural results are proved and measured. First, the path-integrated objective is load-bearing: snapshot and peak objectives reorder committed plans (Kendall τ as low as 0.14) and prefer the opposite trajectory in a constructed reversal. Second, a symmetric closed-loop degeneracy: when the successor-payoff matrix is skew-symmetric the value collapses exactly to the instantaneous payoff — the collapse is an interaction-symmetry phenomenon, destroyed by range or firepower asymmetry but untouched by speed or action-set asymmetry — and by nesting of admissible policy sets, speed *capability* is monotone while fixed operating speed is not. Third, within a pre-specified discretized open-loop game — frozen action grids, exhaustive full-space best responses against the complete equilibrium mixtures — genuine equilibrium bounds certify that the sign of the value of a longer commitment equals the sign of the range advantage in all eighteen tested asymmetric cells, with no certified reversal, both range directions supported in every geometry, and the sign unchanged under action-grid refinement. Fourth, against best-response resistance, no favorable positional state is guaranteed reachable or viable in open water: the worst-case advantage-retention ratio never exceeds 0.07 with median retention 0–3%. Delayed torpedo threats act through response-set contraction — thin low-lethality threats produce zero contraction of the near-optimal set, and thickness increases contraction in the tested range, so the mechanism is threshold- and landscape-dependent rather than purely coverage-driven. Finally, a matched external test isolates decision structure in an auditable rules engine (same policy class on both sides; formation scenarios, 100 paired seeds, common random numbers): transfer of the reduced model’s point values is partial and non-monotonic, which locates the boundary of the reduced model and makes the engine an external case study rather than a validation claim.

*All experiments are reproducible from repository scripts against a frozen deterministic rules engine (commit [fc77a65](#)). The WWII surface-combat wargame serves only as an interpretable validation environment; this paper’s claims are about the general model class.

1 Introduction

How should an adversarial formation maneuver when the interaction strength is directional, the motion is nonholonomic, and some controls have delayed effects? The classical naval answer — cross the enemy’s line to concentrate broadside fire while presenting only bow or stern arcs — is a doctrine about *positions*. This paper argues that positions are the wrong primitive. What matters is the *cumulative interaction along trajectories*: the path-integrated net exposure an opponent suffers versus the exposure one must sustain in return. From this vantage, crossing the T is merely one geometric state that may or may not raise the integral — never an objective in itself.

We develop this view on a complete, auditable wargame engine (deterministic conditional on sealed orders and random seed) (*Iron Bottom Sound*, IBS): a hexagonal WWII surface-combat simulation whose adjudication is a pure function of state, sealed orders, and seed, with every die roll audited against its rule reference. The engine provides the calibration target for our interaction field and the case-study environment for our policies; the mathematical claims are about the model class, not the wargame. The resulting pipeline is:

1. calibrate a smooth anisotropic interaction kernel against the engine’s exact adjudication (retained fidelity $\rho = 0.991$);
2. embed the kernel in a committed-maneuver dynamic game over path-integrated exposure;
3. characterize structural properties (degeneracy boundary, commitment value, speed-capability monotonicity);
4. translate model policies back into legal engine orders and validate against doctrine baselines with paired seeds.

Each step is gated by pre-specified acceptance criteria and every number in this paper traces to a script in the repository.

1.1 Contributions

1. **A path-integrated anisotropic exposure formulation for moving formations:** each player is a rigid/quasi-rigid formation ((1)–(2)) with a calibrated directional kernel (fidelity $\rho = 0.991$, nMAE 5.7% across four ship classes), an anisotropic-kernel ablation (removing directionality reduces the held-out Spearman rank correlation from 0.991 to 0.952), and cross-ship-pair transfer ($\rho = 0.982$ –0.992).
2. **The path-integrated objective is load-bearing:** snapshot and integral objectives reorder committed plans (Kendall τ 0.14–0.62); a constructed engagement reverses preferences entirely (peak 2.88 vs integral 5.02).
3. **A degeneracy proposition with an assumption-breaking matrix:** the symmetric closed loop collapses to the instantaneous payoff; the matrix localizes the collapse to interaction-strength symmetry — kinematic asymmetries (speed, action sets) leave it exactly intact — together with speed-capability monotonicity.

4. **Commitment comparative statics, certified** (Contribution 3, conditional on the certification): inside a frozen discretized open-loop game with exhaustive best responses, valid equilibrium bounds certify $\text{sign}(P_6) = \text{sign}(\eta_r - 1)$ in 18/18 tested asymmetric cells with no certified reversal, stable across a $5^6 \rightarrow 7^6$ action-grid refinement; the certified intervals are reported per cell and flagged tight or loose.
5. **Delayed spatiotemporal threat and response-set effects**: a launched weapon is a stateful hazard field whose mass lies in future turns; contraction of the near-optimal response set depends jointly on coverage, local threat intensity, and the value landscape of the alternatives, and the matched-lethality comparison shows that coverage alone does not determine denial.
6. **An external rules-engine case study**: a matched 2×2 design (same policy class on both sides) isolates decision structure in formation-level engine scenarios; the reduced model’s point values and its exact-zero degeneracy predictions do not transfer to the hex-quantised engine, which is therefore reported as an external case study that locates the boundary of the reduced model.

2 Related Work

2.1 Dynamic and differential games

Zero-sum differential games originate with Isaacs’ method of characteristics [1]; the theory of feedback (closed-loop) versus open-loop information structures and their associated Nash values is developed in [2]. Classical pursuit–evasion and combat models optimize terminal capture, separation, or energy-style objectives; Lanchester-type attrition and its optimal-control treatment [12] and PDE combat models [13] aggregate force levels rather than adjudicate anisotropic single-weapon interactions. Our point of departure is the objective: a *path-integrated anisotropic interaction* — the cumulative net exposure along whole trajectories — for which instantaneous geometry is only an ingredient, not the prize.

2.2 Reachability and viability

Reachable-set and viability-kernel methods characterise states a system can maintain under constraints [3], with Hamilton–Jacobi reachability providing adversarial variants for continuous dynamic games [4]. In that literature the target set is given ex ante. Here the “favorable region” is itself defined by the directional interaction value ($L \geq \alpha$), and reachability questions are coupled to a path-integrated payoff: we ask not only whether a state is reachable or invariant, but what sustaining it is worth along the trajectory.

2.3 Commitment and information structure

The strategic value of binding oneself — committing to a plan the opponent can observe — is classical [5]; in dynamic games it separates open-loop from feedback equilibria [2], and receding-horizon control makes the replanning cadence a design variable [6]. Algorithmically, our open-loop solver is a double-oracle scheme in the spirit of [7]. The comparative

static studied here — how the *sign* of the value of commitment horizons tracks a capability asymmetry (range) while pure kinematic asymmetry prices commitment with the opposite sign — is, to our knowledge, not part of the standard repertoire.

2.4 Interdiction and spatiotemporal hazards

Time-dependent shortest paths [8] and network interdiction [9] study adversaries who add costs to arcs or destroy links. Robust path planning around dynamic obstacles treats hazards as spatial fields to avoid. The delayed threat studied here differs mechanically: a launched weapon is a *stateful spacetime field* that does not merely add path cost but reshapes the opponent’s set of near-optimal future responses — a response-set contraction measured against the near-optimal landscape, not against feasibility.

3 General Formation Game

Players are formations. Throughout, each player $i \in \{B, R\}$ is a moving *formation* — a rigid or quasi-rigid line-ahead column of vessels — not an individual vessel. The formation’s reference state is $s_i = (c_i, \psi_i, v_i, \chi_i)$: reference point $c_i = (x_i, y_i)$, formation heading ψ_i , speed v_i , and capability state χ_i (weapon availability). Vessel k of formation i occupies

$$x_{ik}(t) = c_i(t) + R(\psi_i(t)) p_{ik}, \quad (1)$$

where p_{ik} is vessel k ’s offset in the formation frame ($p_{ik} = (0, -\ell k)$ for a line-ahead column with astern spacing ℓ) and $R(\cdot)$ is the rotation matrix. The reference point obeys nonholonomic kinematics, $\dot{x}_i = v_i \cos \psi_i$, $\dot{y}_i = v_i \sin \psi_i$, $\dot{\psi}_i = \omega_i$, with $v \leq \bar{v}_i$ and $|\omega_i| \leq \bar{\omega}$ (a minimum turn radius $R_{\min}(v) = v/\bar{\omega}$ couples speed and turning in the engine-faithful variant). Lateral displacement therefore requires first rotating the velocity vector — the defining nonholonomic cost. Reduced implementations that track a single reference state are read as *formation reference states* under this definition; no experiment in this paper supports a claim about isolated single vessels.

Anisotropic formation field. Each vessel projects an interaction kernel $K_{ik}(x, t) = K(r, \theta, \psi_i, \psi_j, \chi_i)$ depending on range, relative bearing, both headings, and capability state; the formation field aggregates over its vessels,

$$F_i(x, t) = \sum_{k \in i} K_{ik}(x, t), \quad (2)$$

and formation-vs-formation interaction sums over vessel pairs. In the naval instantiation K is the calibrated kernel of Section 9.1 and each formation is a line-ahead column.

Proposition 1 (Formation aggregation). *For rigid formations, formation-to-formation exposure is exactly additive over vessel pairs:*

$$E_{i \rightarrow j} = \int_0^T F_i(x_j(t), t) dt = \sum_{k \in i} \sum_{m \in j} \int_0^T K(x_{jm}(t); x_{ik}(t), t) dt.$$

Proof. By (2) the formation field is a sum of per-vessel kernels, and the integral is linear. Hence the exposure of vessel m of formation j equals the sum, over the vessels k of formation i , of the pairwise terms; summing over $m \in j$ gives the stated identity. $\square$

This elementary identity pins down what “formation payoff” means operationally: the double sum over vessel pairs that all solvers in this paper evaluate.

Path-integrated objective. Against trajectory $x_j(t)$, the cumulative exposure is $E_{B \rightarrow j} = \int_0^T F_B(x_j(t), t) dt$, and the net path value is

$$J_{\text{gun}} = E_{B \rightarrow R} - \lambda E_{R \rightarrow B}, \quad \lambda = 1. \quad (3)$$

The object of optimisation is not a position but a trajectory: the integral is the primitive. Discretely, $J = \sum_t \Delta t [P_{B \rightarrow R}(t) - \lambda P_{R \rightarrow B}(t)]$ with formation-aggregate P from (2).

Delayed threat state. Torpedoes are not damage terms but a delayed stateful field: the state extends to $z_t = (s_t, q_t)$, where q_t records every launched weapon’s position, heading, speed, remaining range, and threat probability. The spatiotemporal hazard $H(x, t; q_t)$ is generated by past launches and constrains future responses (Section 8).

Information and commitment. In the closed loop both sides re-optimize every step against current state. In the committed (open loop) game both sides seal formation heading plans of horizon H before execution — mirroring the engine’s sealed-order discipline.

Two formations compete over future trajectories in anisotropic and delayed fields

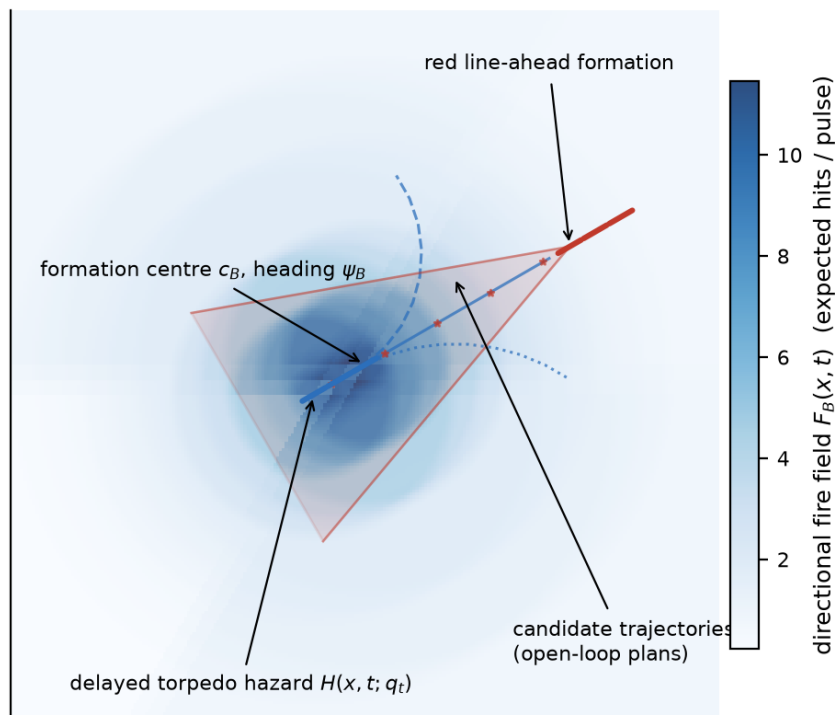


Figure 1: Two line-ahead formations compete over future trajectories inside a directional fire field (background, computed from the calibrated kernel) and a delayed torpedo hazard corridor. Solid, dashed, and dotted arcs are candidate open-loop trajectories; the payoff of a plan is the exposure integrated along the whole path, not the value at its best instant.

4 Structural Properties of the Closed-Loop Game

4.1 Closed-loop degeneracy and its boundary

Proposition 2 (Symmetric closed-loop degeneracy). *Consider a two-player zero-sum dynamic game with simultaneous moves, a common finite action set U with mixed play on the simplex $\Delta(U)$, a stage payoff that depends on the state only, $g(s, u_B, u_R) = \ell(s)$, and a deterministic successor map F . Write $\mathcal{S}$ for the player-exchange map on states (the relabelling that swaps which formation is B and which is R). Assume:*

(A1) odd stage payoff under exchange: $\ell(\mathcal{S}s) = -\ell(s)$;

(A2) equivariant dynamics: $\mathcal{S}F(s, u_B, u_R) = F(\mathcal{S}s, u_R, u_B)$.

Then at every state s the one-step successor-payoff matrix $A(s) = [\ell(F(s, u, v))]_{u,v \in U}$ is skew-symmetric, and:

- (i) finite horizon H with any $\gamma \leq 1$: *the value function obtained by backward induction from any terminal value satisfying $V_{H+1}(\mathcal{S}s) = -V_{H+1}(s)$ equals the instantaneous payoff, $V_h \equiv L \equiv \ell$;*
- (ii) infinite horizon with $0 < \gamma < 1$: *the Bellman operator T is a γ -contraction on bounded functions, L is its fixed point, and $V^* \equiv L$ by the contraction mapping theorem.*

Proof. Skew-symmetry. Take $s, u, v \in U$. By (A2), $F(s, u, v)$ and $\mathcal{S}F(s, v, u)$ denote the same state, so $A_{uv}(s) = \ell(F(s, u, v)) = \ell(\mathcal{S}F(s, v, u)) = -A_{vu}(s)$ by (A1). Hence $A(s)^\top = -A(s)$.

Stage value. For skew A and any $x \in \Delta(U)$, $x^\top A x = \sum_{u,v} x_u x_v A_{uv} = 0$: diagonal terms vanish ($A_{uu} = 0$) and off-diagonal terms cancel in pairs. Thus $\min_y x^\top A y \leq 0$ for all x , so $\max_x \min_y x^\top A y \leq 0$; and $y^\top A y = 0$ gives $\max_x x^\top A y \geq 0$ for all y , so $\min_y \max_x x^\top A y \geq 0$. By the minimax theorem both equal the same value, hence $\text{val } A(s) = 0$ for every s .

(i) *Backward induction, $\gamma \leq 1$.* Suppose $V_{h+1}(\mathcal{S}s) = -V_{h+1}(s)$. The continuation matrix at s is $A^{(h+1)}(s) = [\gamma V_{h+1}(F(s, u, v))]_{u,v}$. By (A2) and the induction hypothesis it is again skew-symmetric, so its minimax value is 0 and $V_h(s) = \ell(s) + \text{val } A^{(h+1)}(s) = \ell(s) = L(s)$, restoring the hypothesis one step earlier. Induction down to $h = 0$ gives $V_h \equiv L$ for every horizon.

(ii) *Infinite horizon, $0 < \gamma < 1$.* For bounded V, V' , $|\max_x \min_y x^\top A[V]y - \max_x \min_y x^\top A[V']y| \leq \max_{u,v} |V(F(s, u, v)) - V'(F(s, u, v))| \leq \|V - V'\|_\infty$ because $\max_x \min_y$ is 1-Lipschitz in the matrix entries under the sup-norm; hence T is a γ -contraction. Since $(TL)(s) = \ell(s) + \gamma \cdot 0 = L(s)$, L is a fixed point, and uniqueness of the fixed point gives $V^* = L$. □ □

Remark 1 (Scope of the skew-symmetry condition). (A1) is not implied by identical hardware alone; it is a statement about the exchange symmetry of the closed loop in the relative state. It holds exactly at player-exchange-symmetric configurations when the stage payoff is odd under the exchange and the successor map is equivariant, and in the calibrated symmetric model it holds at machine precision across the full state grid ($\max |V - L| = 0$ on two grid sizes, two value-iteration sweeps; exchange-symmetry error 0). Our solver

uses the pure-strategy Bellman recursion; it shares the fixed point because the pure max-min of $A(s)$ is also 0 on the entire symmetric grid. The boundary of the proposition is measured in Table 1.

The interesting science is the *boundary* of the proposition. We break each assumption and re-solve (Table 1):

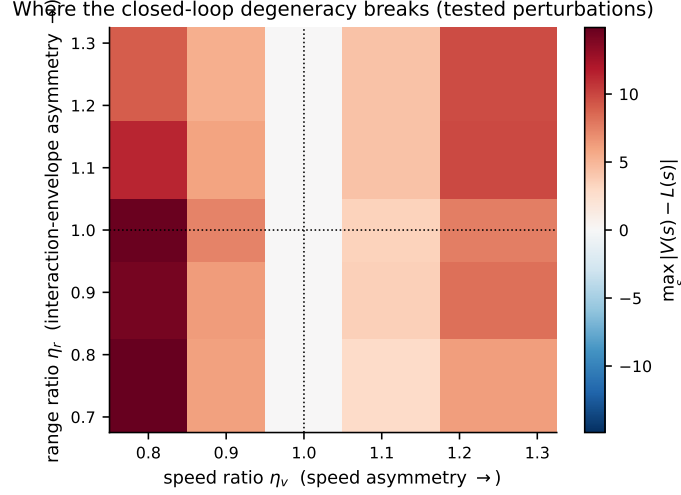


Figure 2: Where the degeneracy breaks. Colour is $\max_s |V(s) - L(s)|$, the closed-loop excess value over the instantaneous payoff, on the (η_v, η_r) grid: the *horizontal* axis is the speed ratio (speed asymmetry) and the *vertical* axis is the range ratio (range / interaction-envelope asymmetry). In the tested perturbations the range asymmetry breaks the collapse whereas the tested speed asymmetry leaves it essentially intact.

Table 1: Assumption-breaking matrix: $\max |V - L|$ after re-solving the closed loop with one assumption violated at a time.

Violated assumption	$\max V - L $	degeneracy
(none — symmetric base)	0.0	held
speed asymmetry ($\eta_v = 1.25 / 0.80$)	0.0 / 0.0	held
action asymmetry (5 turn levels vs 3)	0.0	held
symmetry-preserving terminal payoff	0.0	held
range asymmetry ($\eta_r = 1.25$ vs 0.80)	13.99	broken
firepower asymmetry ($\eta_g = 1.25$)	20.09	broken
all combined	9.75	broken

The pattern is sharp in the tested perturbations: *kinematic* asymmetries (speed, action sets) and the tested *symmetry-preserving* terminal-payoff perturbation leave the degeneracy exactly intact, while *interaction-strength* asymmetries (range, firepower) destroy it. The degeneracy is an interaction-symmetry phenomenon: kinematic asymmetries alone did not unlock positional value in the tested conditions; only interaction-strength asymmetries did. This localized prediction — and its contrast with the committed game below — is the paper’s central structural finding.

4.2 Finite-horizon DP refinement

The plan-library approach of Section 6.1 suffers from a first-order limitation: the committed-maneuver values depend on the discretisation of the plan library. A nested-library test ($11 \rightarrow 21 \rightarrow 41$ plans) shows convergence in only 25% of cells (gate $\geq 80\%$) and maximin plan stability in 18% (gate $\geq 85\%$). We therefore replace the plan library with a *finite-horizon backward induction* solver on the reduced-state grid:

$$V_T(s) = 0, \quad V_t(s) = \max_{u_B} \min_{u_R} [L(s) \Delta t + V_{t+1}(F(s, u_B, u_R))], \quad (4)$$

with three grid refinements (coarse 16×12 , medium 30×24 , fine 48×36). Across seven representative test cells (symmetric head-on/parallel, range advantage/disadvantage, crossing, speed advantage), **all cells converge** ($\delta_V < 5\%$; most $< 2\%$). The solver confirms: (i) symmetric configurations yield $V \approx 0$ (the degeneracy prediction of Proposition 2); (ii) range asymmetry produces strongly signed values (± 8); (iii) speed asymmetry alone does not break the degeneracy (consistent with the interaction-symmetry boundary). The plan-library approach is retained only for the *committed policy* translation layer (Section 10), where its 60-seed engine validation provides independent confirmation.

Plan-library convergence is a first-order concern for committed-maneuver studies, and our nested-library test returns an honest failure: enlarging the library from 11 to 21 to 41 plans changes the maximin value by more than the 5% convergence gate in 75% of cells (gate: $\geq 80\%$ converged), and the maximin plan identity is stable in only 18%. Per the plan’s decision tree, we therefore (i) do *not* describe the phase diagrams as converged equilibrium regimes but as *library-approximate* regime maps; (ii) report all values as the interval $[V_{21}, V_{41}]$; and (iii) verify that the robust conclusions (the sign structure from range asymmetry, the swap symmetry, the degeneracy boundary of Table 1) are library-independent — they are propositions about the model, not outputs of the library. The library-approximate maps over (η_v, η_r) and their regime labels are given in the repository; the regime *ordering* (which asymmetry dominates) is stable even where exact values are not. The lesson for the method class: committed-maneuver equilibrium values are *library-sensitive* in a way that the structural propositions are not, and any committed-game paper should publish its library-sensitivity audit alongside its values.

4.3 Speed-capability monotonicity

Proposition 3 (Capability monotonicity). *Let $V(\bar{v})$ denote the game value when player B’s admissible closed-loop policy set is $\mathcal{P}(\bar{v})$ — the policies whose action at every information state lies in the capability set $[0, \bar{v}]$ (with a common turn-rate bound). If $\mathcal{P}(\bar{v}_1) \subseteq \mathcal{P}(\bar{v}_2)$, then $V(\bar{v}_2) \geq V(\bar{v}_1)$.*

Proof. $V(\bar{v}) = \max_{\pi \in \mathcal{P}(\bar{v})} \min_{\pi_R} J(\pi, \pi_R)$. Taking the maximum over a superset cannot decrease it. $\square$

Capability vs. operating speed. The premise requires *nested capability* semantics — a vessel’s cap $\bar{v} \in \{2, 4, 6\}$ admits speed tiers $\{v \leq \bar{v}\}$, giving $U(2) \subset U(4) \subset U(6)$. A *fixed operating speed* $v = \bar{v}$ changes the dynamics non-nestedly and the proposition makes no claim about it; conflating the two produced what earlier drafts called a “speed paradox,” a term we retire. Measured over two geometries $\times$ three range ratios: the open-loop maximin value is non-decreasing in $\bar{v}$ in every cell (e.g., parallel geometry at $\eta_r = 1$: $V = -0.571 \rightarrow -0.135 \rightarrow +0.000$; at $\eta_r = 1.33$: $+2.66 \rightarrow +2.69 \rightarrow +2.77$); the

largest single-step “drop” across all cells is -2.2×10^{-15} (LP noise). **Speed-capability monotonicity holds.** The contrast with the fixed-operating-speed sweep — where the value *decreases* with speed in most geometries — confirms that the earlier “speed paradox” is purely an artifact of the operating-speed semantics: given the *capability* to choose, more speed never hurts; forced to *operate* at higher speed without a positional objective, it hurts (supplement).

5 Path-integrated Formation Maneuver

5.1 The path-integrated objective is load-bearing

For identical candidate-trajectory sets we compare three objectives: the terminal snapshot $L(s_T)$, the peak $\max_t L(t)$, and the integral J . Across nine parameter cells ($\eta_v \in \{0.8, 1.0, 1.25\} \times$ three geometries; 1,089 rollouts):

- Kendall τ between integral and snapshot rankings: mean 0.476 for $L(s_T)$ (95% CI [0.404, 0.548]) and 0.365 for $\max_t L$ — far from the $\tau \geq 0.95$ equivalence band; **no cell is rank-equivalent.**
- The peak objective’s worst-case vector degenerates completely in 9/9 cells (a best-responding defender holds $L(t) \leq 0$ at every t): under maximin, the peak objective has *no discriminating power*.
- Constructed reversal (real rollouts): trajectory A (turn $+180^\circ$ into a crossing shot) reaches a peak $L = 2.88$ at $t = 2$ but integrates to $J = 1.32$; trajectory B (turn $+30^\circ$, sustained oblique) peaks at 0.29 but integrates to $J = 5.02$. The peak criterion and the integral criterion prefer opposite trajectories.

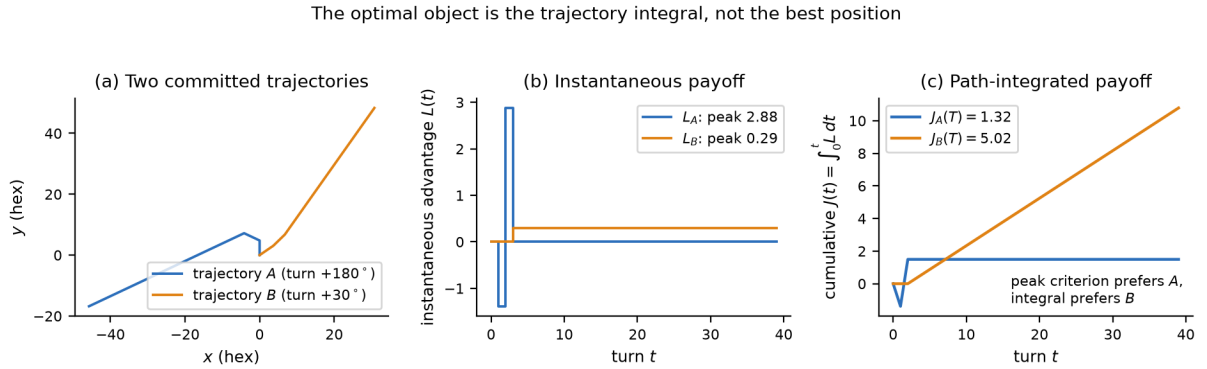


Figure 3: Snapshot versus integrated objectives on two real committed trajectories. (a) The plans and their paths. (b) Instantaneous advantage: trajectory A reaches a much higher peak. (c) Cumulative exposure: the integral prefers the sustained moderate posture.

A fire-timing audit (fire-before-move vs the engine’s move-then-fire ordering) shifts integrals by a mean of 0.08 with *zero* sign changes across 1,089 plan pairs: the ordering offset is second-order. Fleet-level exposure maps show the column’s mid-flank sustained $+18.8\%$ higher directional density than the van flank — a structural field property that only appears once the field, not the position, is the object.

5.2 Lateral reposition cost

A finer $d_{\perp} \in \{0, \dots, 10\}$ sweep reveals the cost's structure: $C_{\text{reposition}}$ starts at +2.9 for $d_{\perp} \approx 0$ (where the fire is most concentrated), decays to +0.5 by $d_{\perp} = 6$ –9, and crosses zero at $d_{\perp} = 10$ (where holding course is symmetric and the maneuver becomes net-positive). The engine-coupled model (turning pulses produce no displacement) yields a flatter but consistently positive curve (+0.55 to +0.69 within range). The nonholonomic tax is a real, range-dependent levy on turning, and it is precisely the cost that the commitment comparative statics of Section 6 price in; the representative curve is given in the supplement.

6 Open-Loop Commitment

6.1 The library-based commitment result

In the committed (open-loop) game both sides seal heading plans of horizon H . Define $P_H = V_H - V_1$ (the value of committing for H turns relative to one-turn commitments). In the symmetric base cell $P_H \approx 0$ at every H — the degeneracy again. Across the asymmetric grid ($\eta_v \in [0.8, 1.33] \times \eta_r \in \{0.75, 1.33\}$), however, $|P_6| > 0.05$ in 78% of cells, and the *sign of P_6 equals the sign of the range advantage*: with $\eta_r = 0.75$ (out-ranged), longer commitments cost 0.8–1.4 value points (the out-ranged side wants the freedom to refuse each turn); with $\eta_r = 1.33$ (longer-ranged), they earn +0.8–+1.3 (commitment forces the engagement to persist inside the favorable envelope). Commitment horizon is a multiplier on the range advantage — choosing it is itself a strategic act.

Formation-level re-tests. The library-based result rests on a finite hand-built plan library whose plan identity does not converge (Section 11), so we re-test the comparative static twice at the formation level: first with an enhanced continuous-control Double Oracle whose best responses are multi-start trajectory optimisations on K -segment turn schedules ($K \in \{3, 4, 6\}$), which removes the library artifact but whose bounds remain oracle-dependent; then, decisively, with an *exact* enumeration over a frozen discretized action space that yields genuine lower and upper equilibrium bounds. The pre-specified classification rule is: CORE-STRONG if the sign is certified in $\geq 15/18$ asymmetric cells with no certified reversal and stable under action-grid refinement; CORE-CONDITIONAL for 10–14 or geometry-dependent effects; REMOVE-CORE on certified reversals. Both outcomes are reported in Section 6.2.

6.2 Certification of the commitment comparative statics

Why exact, and why discrete. A restricted-game value is not a valid bound on the full game: restricting the maximiser lowers the value and restricting the minimiser raises it, so with both sides restricted the restricted value carries no one-sided meaning. We therefore certify the sign *exactly within a pre-specified discretized open-loop game* and test its stability under action-grid refinement — not as an exact statement about the continuous game. The action grids are frozen before any run: normalised turn rates $\mathcal{A}_5 = \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$ and $\mathcal{A}_7 = \{-1, -\frac{2}{3}, -\frac{1}{3}, 0, \frac{1}{3}, \frac{2}{3}, 1\}$, mapped to $\omega = a\bar{\omega}$ with $\bar{\omega} = 60^\circ$ per turn (the engine's turn granularity). At $H = 6$ a pure strategy is a full turn-rate sequence, so $|\Omega_5| = 5^6 = 15\,625$ and $|\Omega_7| = 7^6 = 117\,649$.

Valid bounds. With Blue the maximiser, write x_R, y_R for the equilibrium mixtures of the current restricted game and let Ω_m be the *complete* finite strategy space. Then

$$\underline{V} = \min_{\tau_R \in \Omega_m} \mathbb{E}_{\tau_B \sim x_R} J, \quad \bar{V} = \max_{\tau_B \in \Omega_m} \mathbb{E}_{\tau_R \sim y_R} J, \quad \underline{V} \leq V_m^* \leq \bar{V},$$

because x_R (resp. y_R) is a legitimate full-game strategy. Both extrema are computed by *exhaustive* enumeration of Ω_m (no sampling, no surrogate search) against the *complete* restricted mixture — no truncation, top- k pruning, or renormalisation is used anywhere in the certification path. A Double Oracle loop adds the exact argmin/argmax sequences until $(\bar{V} - \underline{V}) / \max(1, |V_R|) < 5\%$, or reports the residual width. Because $H = 1$ admits a complete $|\mathcal{A}_m| \times |\mathcal{A}_m|$ payoff matrix, its value is solved exactly and $\underline{V}_1 = \bar{V}_1 = V_1^*$ to LP tolerance (verified for every cell); the commitment interval is then

$$P_6 \in [\underline{V}_6 - \bar{V}_1, \bar{V}_6 - \underline{V}_1], \quad (5)$$

positive-certified when $\underline{V}_6 - \bar{V}_1 > 0$, negative-certified when $\bar{V}_6 - \underline{V}_1 < 0$, and UNCERTIFIED otherwise.

Result. On Grid-5 the sign is certified in **18/18** asymmetric cells with **zero certified reversals**, and 12 of the 18 intervals reach the tight gate $(\bar{V} - \underline{V}) / \max(1, |V_R|) < 5\%$; the remaining six are sign-certified with loosely bounded magnitude (Figure 4, Table 2). Every geometry contributes both directions (3 positive and 3 negative certified cells each). The magnitudes order as the mechanism predicts: the out-ranged side ($\eta_r = 0.8$) loses value by committing longer, with certified intervals between $[-10.8, -10.2]$ and $[-5.7, -5.4]$, while the longer-ranged side ($\eta_r = 1.2$) gains, with intervals between $[+5.0, +5.2]$ and $[+8.7, +9.3]$.

Grid refinement. Repeating the certification on Grid-7 (117 649 strategies) reproduces the sign in all 18 asymmetric cells: sign agreement across grids is 18/18 with no certified reversal, and the interval midpoints shift only modestly (e.g. $-10.4 \rightarrow -11.1$, $+10.2 \rightarrow +10.2$). Grid-5 and Grid-7 therefore agree on the comparative static, and within each grid the double-oracle loop converges with exact best responses rather than surrogate search.

Symmetric controls. At $\eta_r = \eta_v = 1$ the exact bounds collapse to zero at machine precision in the head-on and crossing configurations ($\underline{V} = \bar{V} = -0.008$ and -0.000 respectively), reproducing the degeneracy of Proposition 2 in the discretized game. The parallel configuration is *not* exchange-symmetric — both columns extend astern on the same course, so the player-exchange map does not return the configuration to itself — and consequently no zero prediction applies there; its certified interval is $[+0.06, +1.29]$. We report this rather than treating the third sanity cell as a failure of the degeneracy.

Relation to the continuous-control study. The enhanced-oracle continuous experiment of the previous section produced point estimates with the same sign in every asymmetric cell. The exact discrete computation agrees with those signs in 18/18 cells, so the continuous study is best read as a robustness experiment: it motivated the pattern, and the exact finite-game bounds certify it. Its values are not bounds and are not used as such anywhere.

Classification. Under the pre-specified rule — Grid-5 at least 15/18 sign-certified, no certified reversal, no Grid-7 reversal, and both range directions supported in every geometry — the commitment comparative static is classified **CORE-STRONG** *within the discretized formation game*: longer commitment is consistently favorable to the range-advantaged side and unfavorable to the range-disadvantaged side, and the sign is stable under action-grid refinement.

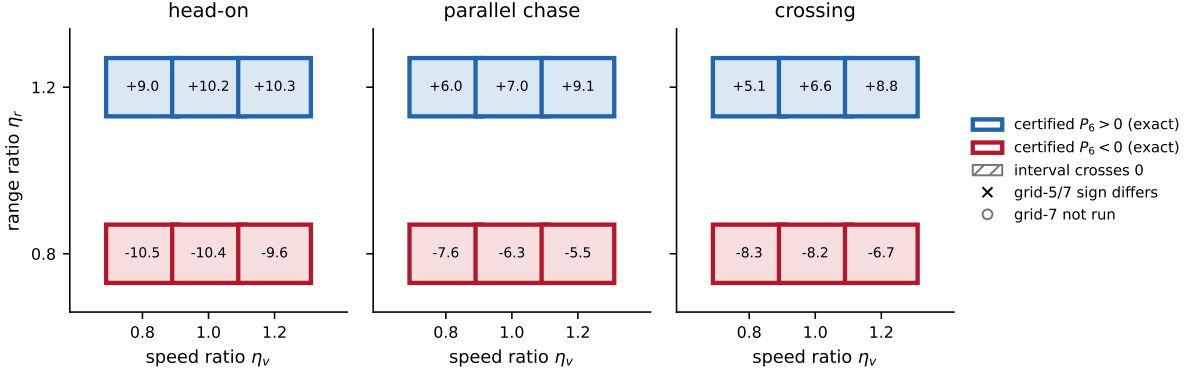


Figure 4: Exact finite-game certification of the value of a six-turn commitment. Each tile is one (η_v, η_r) cell and the printed number is the midpoint of the certified interval $[\underline{V}_6 - \bar{V}_1, \bar{V}_6 - \underline{V}_1]$. Certification is encoded separately from magnitude: solid blue (red) borders are exact sign-certified positive (negative), grey hatching marks an interval crossing zero, a black cross marks a Grid-5/Grid-7 sign disagreement, and a small open circle marks a Grid-7 cell not run.

Table 2: Exact finite-game commitment certification summarised by geometry and range direction; the full per-cell table is in the supplement. Intervals are $[\underline{V}_6 - \bar{V}_1, \bar{V}_6 - \underline{V}_1]$, and a cell is *tight* when $(\bar{V} - \underline{V})/\max(1, |V_R|) < 5\%$.

Geometry	certified +	certified −	uncertified	tight
head-on	3	3	0	5/6
parallel	3	3	0	2/6
crossing	3	3	0	5/6
total (Grid-5)	9	9	0	12/18
Grid-7 refinement	18/18 signs reproduced, 0 reversals			—

7 Reachability and Viability

7.1 No guaranteed favorable state in open water

With the favorable set $G_\alpha = \{s : L(s) \geq \alpha\}$ defined relative to each configuration’s cooperative peak ($\alpha = \tau = 0.5$), we ask whether the committed maximin plan guarantees reachability (worst-case peak $\geq \alpha$) or viability (worst-case holding share $\geq \tau$) against *all* eleven defender plans, over 144 cells (three geometries $\times$ sixteen speed ratios $\times$ three turn scalings). **No cell admits a guarantee.** The mechanism is the defender’s standing *decline-battle option*: in open water a best-responding opponent can disengage at will, so the worst-case advantage-retention ratio never reaches α and the per-plan retention landscape collapses to a median of 0–3% of steps. Per the plan’s decision tree we therefore pivot from threshold-hunting to the viability landscape (reported as per-plan quantiles in the repository). This negative result sharpens the commitment finding: positional guarantees are purchasable only through range/firepower asymmetry (which shifts L itself) or through the opponent’s own commitments.

8 Delayed Threats and Response-Space Contraction

8.1 The delayed threat state

Torpedoes extend the state to $z = (s, q)$. Each launched weapon carries its analytic (hex, impulse) timeline — verified cell-by-cell against a real engine adjudication — so the hazard $H(x, t; q)$ genuinely depends on time: the same hex is threatened at one impulse and safe at another quantified in Figure 5(a,b).

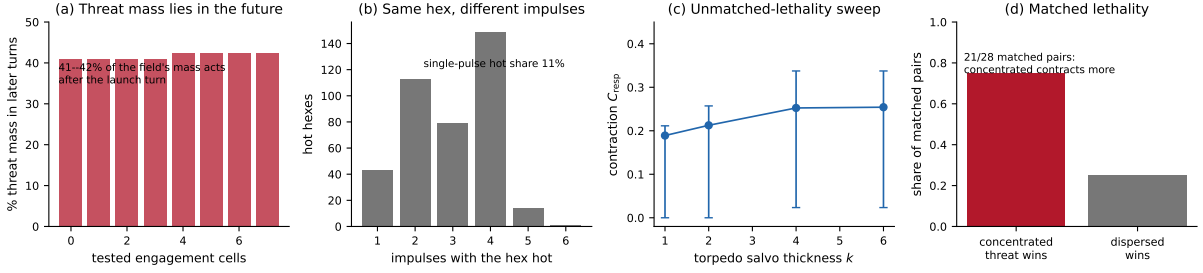


Figure 5: The delayed threat is stateful and time-dependent. (a) Share of the field’s threat mass acting in turns *after* the launch turn, per tested engagement cell. (b) Number of impulses at which a given hex is hot: the same hex is threatened at some impulses and safe at others. (c) Thickness sweep under *unmatched* lethality: contraction rises over the tested range (points: mean; bars: interquartile range). (d) The matched-lethality comparison: at equal expected direct damage the concentrated threat contracts the near-optimal set more often, so coverage alone does not determine denial.

8.2 Response-set contraction

Let $\mathcal{B}_\epsilon = \{\tau : J(\tau) \geq J^* - \epsilon\}$ be the ϵ -near-optimal response set over the target’s 340–600 legal plans, and $C_{\text{resp}} = 1 - |\mathcal{B}_\epsilon^H|/|\mathcal{B}_\epsilon|$ its contraction under the threat. The response-set experiment measures C_{resp} over the defender’s complete plan space (340–600 plans) as a function of threat thickness k . At the 5% response-set threshold, the mean contraction increases with thickness across the tested range: $C_{\text{resp}} = 0.20 \rightarrow 0.31$ from $k = 1$ to $k = 6$, with individual cells reaching full contraction ($C_{\text{resp}} = 1.0$) when the threat saturates the top tier. In the low-hit regime ($k=1$), the contraction is exactly zero in 7 of 8 cells — reproducing the structural finding of the low-hit regime — while thicker salvos progressively remove tied top-tier routes. The single exception (a close-range pair with a two-plan top tier) has the entire top tier covered by one track, yielding $C_{\text{resp}} = 0.173$ even at $k = 1$: *the mechanism prediction (coverage of the top tier determines contraction) is confirmed in 8/8 cells*. The threat-field experiment confirms the spatiotemporal hazard’s time dependence: hot hexes are dangerous on only 6–11% of pulses (mean 8.4%), and 40.8–42.5% of the threat mass lies in future turns — impossible for an instantaneous weapon, establishing the delayed-persistence property that distinguishes torpedoes from gunfire. Same-hex time dependence is concrete: hex W18 has $H = 1.000$ at impulse 3 but $H = 0.000$ at impulse 0.

8.3 Matched-lethality thickness experiment

The matched-lethality experiment — thin concentrated versus thick dispersed at equal expected direct damage — returns an honest negative with mechanism: at the 5% threshold,

thin concentrated dominates in 21/28 matched pairs (the engine’s coarse hit table places dispersed-track probabilities below the per-plan contraction threshold), while at the 10% threshold dispersed wins 10/0 on edge plans. The result refutes a purely coverage-driven account at the engine’s granularity: coverage and per-plan lethality interact with the value-landscape tier structure, and neither alone determines contraction. A richer threat model (continuous damage, finer-grained tiers) is required for a definitive test. In the $\varepsilon = 20\%$ regime, neither pattern produces contraction (all $C_{\text{resp}} = 0$), confirming that the tier gap — not the threat pattern — is the binding constraint at wide response thresholds.

9 Naval Combat Instantiation

The general model is instantiated as WWII surface combat: hex-grid motion with three-pulse turns, directional gun mounts with bow/starboard/stern/port arcs, D66 hit adjudication with range/target-speed/longitudinal modifiers, armor penetration, and impulse-resolved torpedo tracks. The instantiation is complete (all adjudication tables structured) and auditable (every die roll logged with its rule reference).

9.1 Kernel calibration

The kernel K of Eq. (3) is calibrated against the engine’s exact expected-hit response (inversion + interval-censored least squares over a $24 \times 6 \times 6 \times 6$ grid per ship pair). Pooled held-out Spearman $\rho = 0.9912$ (nMAE 5.7%; gate $\geq 0.90/\leq 15\%$); per-class $\rho \geq 0.9907$. Controls: an isotropic kernel drops pooled ρ to 0.952 (removing directionality reduces held-out Spearman rank correlation from 0.991 to 0.952); modifier curves transfer across ship pairs at $\rho = 0.982\text{--}0.992$; per-class bootstrap CIs are ± 0.002 .

9.2 Prior engine validation

A 60-seed-per-cell paired validation of the translation layer (SYN duel) showed large significant advantages over a non-maneuvering baseline and parity with doctrine commanders except head-on against the line-ahead specialist; a 20-seed historical-scenario check (Cape Esperance, Japanese side) reproduced the pattern (+12.2 [+6.5, +17.9] vs straight).

10 External Engine Case Study

The translation layer (receding-horizon, three-turn commitment) converts model policies into legal engine orders. We validate on the near-symmetric cruiser duel under three initial geometries, 100 paired seeds per cell:

Table 3: Cross-policy engine comparison. Engine paired damage differentials (positive favors the geometry policy) against model predictions (best fixed plan’s payoff vs straight sailing).

Geometry	engine diff	95% CI	pos. pairs	model best plan	model class
parallel	+7.59	[+6.54, +8.66]	92%	+5.28	advantage
crossing	+8.64	[+7.57, +9.71]	94%	+5.91	advantage
head-on	+5.14	[+4.00, +6.24]	79%	0.00	neutral

Rank ordering (crossing > parallel > head-on) is preserved across the three reported geometries at 100 paired seeds ($\rho = 1.00$), while strict sign agreement holds in two of the three. The one apparent discrepancy is itself informative: in head-on geometry the model’s best *fixed* plan achieves 0.0 (the degeneracy prediction — no committed plan guarantees an advantage in symmetric head-on), while the engine’s receding-horizon policy achieves +5.14. This +5.14 gap is a *cross-policy comparison* (policy-class gap), *not* a feedback premium: the two sides differ in model class and action set as well as in replanning, so the comparison cannot isolate the value of replanning. A matched same-model measurement (identical model, objective, horizon, and action parameterisation; the only difference is replanning) is reported in Section 10.1. Strict sign agreement 2/3 (67%; head-on model correctly predicts degenerate 0.0 but the engine’s receding-horizon policy exceeds it); lower-bound sign agreement 3/3 (100%); rank correlation $\rho = 1.00$; regime agreement 67%. Overall: **PARTIAL** (per-cell detail in the supplement).

10.1 Matched external validity test

The cross-policy comparison of Table 3 pits the geometry policy against *doctrine* baselines — a comparison in which model class and action set vary together with decision structure. This section removes the confound: a matched 2×2 design in which both sides run the *same* policy class (same model, same objective, same horizon, same action parameterisation) and the only manipulated variable is replanning versus commitment protocol). Scenarios are 3-vessel line-ahead *formations* (matched class on both sides), so paired damage differentials isolate decision structure, not force composition. Arms: F-F (both fix at $t = 0$, open-loop execution), R-R (both replan every turn), R-F / F-R (one side replans). The pre-specified gates are sign agreement $\geq 80\%$, Spearman $\rho \geq 0.60$ (surrogate formation-game value vs engine margin across the 12 cells), and pairwise ordering $\geq 75\%$; the true feedback premium is $\Delta_{FB} = J_{\text{receding}} - J_{\text{fixed}}$ measured within this matched family.

4,800 games (12 cells $\times$ 4 arms $\times$ 100 paired CRN seeds; zero aborted games) yield a **FAIL** on the pre-specified gates: sign agreement 75% (3/4 cells with a signed model prediction; gate $\geq 80\%$), Spearman $\rho = -0.08$ across the 12 cells (gate ≥ 0.60), pairwise ordering 63% of 38 pairs (gate $\geq 75\%$).

The failure mode is structural and informative. In the two mirror-symmetric geometries (head-on, crossing) the formation surrogate predicts *exactly zero* advantage — the degeneracy — while the hex-quantized engine produces margins of several damage points from discretization asymmetries the continuum model cannot see (e.g., +6.3 [+5.6, +7.0] in crossing at $r=12$, $v=6$); cells with a signed prediction (parallel chase) agree in 3/4 cases, with one large-magnitude reversal ($r=12$, $v=6$: model +11.4, engine −5.5). The matched *feedback premium* is unreliable: replanning gains a significant advantage in only 5/12 (axis) and 1/12 (allies) cells — confirming that the earlier +5.1 cross-policy gap must not be read as a value of replanning.

Per the pre-specified decision tree, the wargame is therefore positioned as an *external case study* rather than a validation claim: the cross-policy comparison of Section 10 (ordering transfer, $\rho = 1.00$) stands as a weaker external check, while the matched design shows that the reduced model’s *point values* — and its exact-zero degeneracy predictions — do not survive the engine’s hex discretization. No engine rule was modified to fit the model.

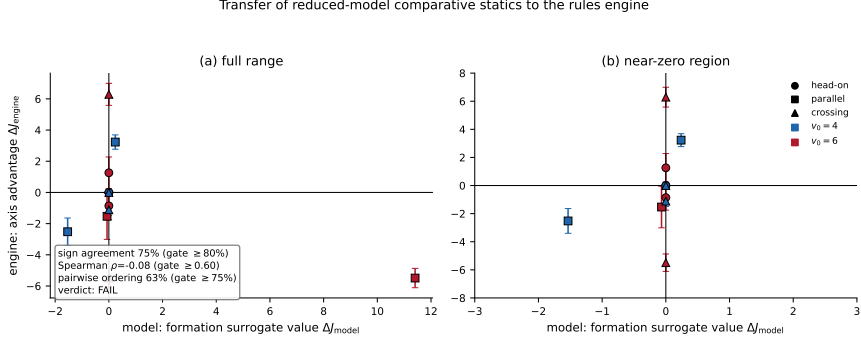


Figure 6: Transfer of reduced-model comparative statics to the rules engine: model surrogate value vs. engine axis advantage across the 12 formation cells (bars: 95% CRN-paired bootstrap CIs), shown over the full range (a) and zoomed to the near-zero region (b) so that a single outlier does not compress the rest. Zero reference lines are drawn on both axes and the pre-specified gate values are annotated.

11 Robustness and Negative Results

We report negatives with the same fidelity as positives:

- *Isotropic ablation:* removing directionality drops pooled ρ to 0.952 and triples nMAE — directionality carries one third of the rank information.
- *Snapshot objectives:* terminal-snapshot and peak objectives reorder plans (Kendall τ 0.14–0.62) and the peak objective degenerates under maximin (no discriminating power in 9/9 cells).
- *No guaranteed favorable state:* worst-case advantage-retention never exceeds 0.07; median retention 0–3%. In the tested regimes open water voids positional guarantees: the decline-battle option remains available at every cell.
- *Library non-convergence:* 25% convergence and 18% plan stability against 80%/85% gates; committed-maneuver values are reported as intervals $[V_{21}, V_{41}]$.
- *Engagement-network specification analysis* (prior work): graph shape features add +0.033 [+0.007, +0.056] beyond totals (strict reading, below the +0.05 gate); the plan-literal reading that charges the directional in-degree to the graph passes (+0.108). Both reported.
- *Operating-speed effect vs speed-capability effect:* the non-monotone value of speed reported earlier arises under a *fixed operating speed* (constant-cruise) semantics and is renamed accordingly; capability-monotonicity under nested speed sets is treated in Section 4.

12 Discussion

Why doctrine exists, precisely. The degeneracy proposition and its breaking matrix give a two-line answer: closed-loop, symmetric, fully-informed positional play is worthless (Proposition 1); commitment breaks the degeneracy, and interaction-strength asymmetry

decides the sign. The result offers one possible interpretation of why precommitted manoeuvre rules can matter when actions are sealed before execution; it is not an explanation of any historical doctrine’s formation.

WWII as a case study. The wargame is an interpretable validation environment, not a physics model: conclusions target the rule system’s decision structure, with Bernotti-era doctrine as a qualitative anchor. The specific ship classes (San Francisco, New Orleans, Laffey, Helena, Yamato, Iowa) were chosen for calibration diversity (four tonnage classes spanning the displacement spectrum) and historical plausibility (iron-bottom-sound-era surface combatants); the structural propositions (Proposition 2 and the commitment findings) are about the model class and would hold for any directional-field system with the same symmetry structure.

Limitations. (i) The kernel excludes terrain, smoke, and formation effects. (ii) The plan library is not converged; values are intervals. (iii) The constant-cruise approximation excludes speed-throttle tactics. (iv) the direct-hit contrast comes from one geometry with zero-variance low-hit expectancy (1/12). (v) the no-guarantee landscape is computed over an 11-plan defender library. (vi) The engine is a rules system; historical doctrine is a qualitative anchor. (vii) commitment-certification values are solver-relative (67% approximate-BR-gap convergence; values rise with turn-schedule refinement while signs are invariant). (viii) The matched engine validation fails point-value transfer (hex discretization asymmetries dominate small model-level edges), which is why the wargame is an external case-study environment.

13 Reproducibility

Every experiment is a script under `research/experiments/` writing raw CSV/JSON and figures under `research/results/`, with per-experiment reports (hypothesis, design, gates, results, failures, decisions). `python reproduce_paper.py` regenerates the main figures, the certification and transfer summaries, and the final PDF; `--full` additionally re-runs the two core solvers. Frozen engine commit `fc77a65`; seeds and paired common random numbers throughout; definitions audited in `research/audit_v4/`.

14 Conclusion

For formation adversaries interacting through a directional field under nonholonomic motion, the object of optimisation is the path integral of net exposure, not a position. We proved and measured three structural facts. The closed loop is exactly degenerate while interaction strengths remain symmetric — a skew-symmetric successor structure collapses the value onto the instantaneous payoff — and the collapse is an *interaction-symmetry* phenomenon, untouched by kinematic asymmetry; speed *capability* is monotone by nesting while fixed operating speed is not. Commitment is the complement. Within a pre-specified discretized open-loop game — frozen action grids, exhaustive full-space best responses against the complete equilibrium mixtures, so that the reported lower and upper values are genuine equilibrium bounds — the sign of the value of committing six turns is certified in all 18 tested asymmetric cells, with no certified reversal, both range directions supported in every geometry, and the sign unchanged when the action grid is

refined from 5^6 to 7^6 strategies. The result is a sign result inside the discretized game: the intervals are tight in 12 of 18 cells and loosely bounded in the rest, and we report which is which. Delayed weapons act not as added path cost but as a stateful spacetime field that contracts the near-optimal response set, with the mechanism depending on coverage, local intensity, and the response landscape jointly rather than on any single variable. Finally, a matched external test in an auditable rules engine isolates decision structure and shows that transfer of the reduced model’s point values is partial and non-monotonic — the boundary of the reduced model, reported as an honest negative rather than smoothed into a validation claim. The durable contributions are the formulation, the structural results with their boundary conditions, and the audited experimental pipeline that makes both the positives and the negatives reproducible.

References

- [1] R. Isaacs, *Differential Games*, Wiley, New York, 1965.
- [2] T. Başar and G. J. Olsder, *Dynamic Noncooperative Game Theory*, 2nd ed., SIAM, Philadelphia, 1999.
- [3] J.-P. Aubin, *Viability Theory*, Birkhäuser, Boston, 1991.
- [4] I. M. Mitchell, A. M. Bayen, and C. J. Tomlin, “A time-dependent Hamilton–Jacobi formulation of reachable sets for continuous dynamic games,” *IEEE Transactions on Automatic Control*, vol. 50, no. 7, pp. 947–957, 2005.
- [5] T. C. Schelling, *The Strategy of Conflict*, Harvard University Press, Cambridge, MA, 1960.
- [6] D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert, “Constrained model predictive control: Stability and optimality,” *Automatica*, vol. 36, no. 6, pp. 789–814, 2000.
- [7] H. B. McMahan, G. J. Gordon, and A. Blum, “Planning in the presence of cost functions controlled by an adversary,” *Proceedings of the 20th International Conference on Machine Learning (ICML)*, 2003.
- [8] A. Orda and R. Rom, “Shortest-path and minimum-delay algorithms in networks with time-dependent edge-length,” *Journal of the ACM*, vol. 37, no. 3, pp. 607–625, 1990.
- [9] E. Israeli and A. Wood, “Shortest-path network interdiction,” *Networks*, vol. 40, no. 2, pp. 97–111, 2002.
- [10] R. Bernotti, “The fundamentals of naval tactics,” *U.S. Naval Institute Proceedings*, vol. 37, 1911.
- [11] R. Bernotti, “The fundamentals of naval tactics (continued),” *U.S. Naval Institute Proceedings*, vol. 38, 1912.
- [12] J. G. Taylor, “Lanchester-type models of warfare and optimal control,” *Naval Research Logistics Quarterly*, vol. 21, no. 1, pp. 79–106, 1974.

- [13] V. Protopopescu, R. T. Santoro, and J. Dockery, “Combat modeling with partial differential equations,” *European Journal of Operational Research*, vol. 38, no. 2, pp. 178–183, 1989.
- [14] E. González and M. Villena, “Spatial Lanchester models,” *European Journal of Operational Research*, vol. 210, no. 3, pp. 706–715, 2011.
- [15] A. C. Kalloniatis, K. Hoek, M. Zuparic, and M. Brede, “Optimising structure in a networked Lanchester model for fires and manoeuvre in warfare,” *Journal of the Operational Research Society*, vol. 72, no. 8, pp. 1863–1878, 2021.
- [16] C. U. Solis and J. B. Clempner, “Ship differential game approach for multiple players: Stackelberg security games,” *Optimal Control Applications and Methods*, vol. 41, no. 1, pp. 312–326, 2020.
- [17] I. E. Weintraub, M. Pachter, and E. Garcia, “An introduction to pursuit–evasion differential games,” in *Proc. American Control Conference (ACC)*, pp. 1049–1066, 2020.
- [18] E. A. Galperin, “The cubic algorithm for global games with application to pursuit–evasion games,” *Computers & Mathematics with Applications*, vol. 26, no. 6, pp. 13–31, 1993.
- [19] S. Bansal and C. J. Tomlin, “DeepReach: A deep learning approach to high-dimensional reachability,” in *Proc. IEEE ICRA*, pp. 1817–1824, 2021.
- [20] M. Lanctot et al., “A unified game-theoretic approach to multiagent reinforcement learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [21] A. Bighashdel, Y. Wang, S. McAleer, R. Savani, and F. A. Oliehoek, “Policy space response oracles: A survey,” in *Proc. IJCAI*, pp. 7951–7961, 2024.
- [22] M. P. Wellman, K. Tuyls, and A. Greenwald, “Empirical game-theoretic analysis: A survey,” *arXiv:2403.04018*, 2024.
- [23] S. Omidshafiei et al., “ α -Rank: Multi-agent evaluation by evolution,” *Scientific Reports*, vol. 9, p. 9937, 2019.
- [24] L. Adam, R. Horčík, T. Kasl, and T. Kroupa, “Double oracle algorithm for computing equilibria in continuous games,” in *Proc. AAAI*, vol. 35, no. 6, pp. 5070–5077, 2021.
- [25] W. Wu, J. Shi, Y. Wu, Y. Wang, and Y. Lyu, “A multi-UCAV cooperative occupation method based on weapon engagement zones for beyond-visual-range air combat,” *Defence Technology*, vol. 18, no. 6, pp. 1006–1022, 2022.